Last updated on June 24, 2026

Shijun ZHANG

Assistant Professor

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Appointments

Assistant Professor (Tenure-Track), Jul 2024 ∼ Present
Hong Kong Polytechnic University (PolyU), Hong Kong SAR, China

Phillip Griffiths Assistant Research Professor, Aug 2022 ∼ Jun 2024
Duke University, United States
Mentors: **Jianfeng Lu** and **Hongkai Zhao**

Research Fellow, National University of Singapore, Singapore Jan 2021 ∼ Jul 2022
Mentor: **Zuowei Shen**

Education

Ph.D. in Mathematics, National University of Singapore, Singapore Aug 2016 ∼ Jan 2021
Thesis: *Deep neural network approximation via function compositions* [PDF, URL]
Supervisors: **Zuowei Shen** and **Haizhao Yang**

B.S. in Mathematics, Wuhan University, China Sep 2012 ∼ Jun 2016
Thesis supervisor: **Xiliang Lv**

Awards

The 2026 Frontiers of Science Award (Mathematics), International Congress for Basic Science (ICBS 2026), URL.

Departmental Best Paper Award, AMA, PolyU, 2024-2025, URL.

The EASIAM (East Asia section of SIAM) Student Paper Prize (First Prize), 2020-2021, URL.

Teaching

Operations Research Methods (AMA 3820), PolyU 2025/26 Sem1, 2025/26 Sem2
Instructor, **Syllabus**

Optimization Methods (AMA 4850), PolyU 2024/25 Sem2, 2025/26 Sem2
Instructor, **Syllabus**

Mathematical Numerical Analysis (Math 361S), Duke University Spring 2024
Instructor, **Syllabus**

Matrices and Vectors (Math 218D-2), Duke University Fall 2023
Teaching assistant

Publications

[number] Author(s). *Paper title.* Journal or conference reference. [Links]

* Corresponding author † Equal contribution

Preprint(s)

[21] Baicheng Li, Haizhao Yang, **Shijun Zhang**. *Sobolev Approximation by Fixed-Size Neural Networks with Arbitrary Accuracy.* Submitted. [arXiv]

[20] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Layer-wise Geometric Approximation Rates for Deep Networks.* Submitted. [arXiv]

[19] Zeyu Li, **Shijun Zhang**, Tieyong Zeng, Feng-Lei Fan. *Neural Network Approximation: A View from Polytope Decomposition.* Submitted. [arXiv]

[18] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Multigrade neural network approximation.* Submitted. [arXiv]

[17] Hanyu Pei, Jing-Xiao Liao, Qibin Zhao, Ting Gao, **Shijun Zhang**, Xiaoge Zhang, Feng-Lei Fan. *NeuronSeek: On Stability and Expressivity of Task-driven Neurons* Submitted. [arXiv]

Published (Accepted)

[16] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Fourier Multi-Component and Multi-Layer Neural Networks: Unlocking High-Frequency Potential.* Neural Networks, DOI:10.1016/j.neunet.2026.109268, to appear. [arXiv, Journal]

[15] Fenglei Fan, Juntong Fan, Dayang Wang, Jingbo Zhang, Zelin Dong, **Shijun Zhang**, Ge Wang, Tieyong Zeng. *Hyper-Compression: Model Compression via Hyperfunction.* IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), DOI: 10.1109/TPAMI.2026.3673772, to appear. [arXiv, Journal]

[14] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Structured and balanced multi-component and multi-layer neural networks.* **SIAM Journal on Scientific Computing**, 47(5):C1059-C1090, 2025. [arXiv, Journal]

[13] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Why shallow networks struggle to approximate and learn high frequencies.* **Information and Inference: A journal of the IMA**, Volume 14, Issue 3, September 2025, iaaf022, https://doi.org/10.1093/imaiai/iaaf022. [arXiv, Journal]

[12] Qianchao Wang†, **Shijun Zhang**†, Dong Zeng, Zhaoheng Xie, Hengtao Guo, Feng-Lei Fan, Tieyong Zeng. *Don't Fear Peculiar Activation Functions: EUAF and Beyond.* **Neural Networks**, 186, June 2025. [arXiv, Journal]

[11] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *Deep network approximation: Beyond ReLU to diverse activation functions.* **Journal of Machine Learning Research**, 25(35):1–39, 2024. [arXiv, Journal]

[10] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *On enhancing expressive power via compositions of single fixed-size ReLU network.* Proceedings of the 40th International Conference on Machine Learning (**ICML 2023**), PMLR 202:41452–41487, 2023. [arXiv, Poster, Conference]

[9] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Neural network architecture beyond width and depth.* Advances in Neural Information Processing Systems (**NeurIPS 2022**), 35:5669–5681, 2022. [arXiv, Poster, Conference]

[8] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation in terms of intrinsic parameters.* Proceedings of the 39th International Conference on Machine Learning (**ICML 2022**), PMLR 162:19909–19934, 2022. [arXiv, Spotlight, Conference]

[7] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation: achieving arbitrary accuracy with fixed number of neurons.* **Journal of Machine Learning Research**, Volume 23, Issue 276, September 2022, Pages 1–60. [arXiv, Journal]

[6] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Optimal approximation rate of ReLU networks in terms of width and depth.* **Journal de Mathématiques Pures et Appliquées**, Volume 157, January 2022, Pages 101–135. [arXiv, Journal]

[5] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Neural network approximation: Three hidden layers are enough.* **Neural Networks**, Volume 141, September 2021, Pages 160–173. [arXiv, Journal]

[4] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network with approximation error being reciprocal of width to power of square root of depth.* **Neural Computation**, Volume 33, Issue 4, April 2021, Pages 1005–1036. [arXiv, Journal]

[3] Jianfeng Lu, Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation for smooth functions.* **SIAM Journal on Mathematical Analysis**, Volume 53, Issue 5, September 2021, Pages 5465–5506. [arXiv, Journal]

[2] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation characterized by number of nuerons.* **Communications in Computational Physics**, Volume 28, Issue 5, November 2020, Pages 1768–1811. [arXiv, Journal]

[1] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Nonlinear approximation via compositions.* **Neural Networks**, Volume 119, November 2019, Pages 74–84. [arXiv, Journal]